APPROXIMATE QUERY ENGINE

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PROBLEMS TO TACKLE

01

SPEED VS. ACCURACY TRADE-OFF

Business users often need fast trends, not exact numbers. AQP delivers insights quickly with acceptable error margins.

02

SAMPLING-BASED EXECUTION

Use techniques like block-level sampling to reduce data scanned and speed up query response.

CHALLENGES

01

SAMPLING ACCURACY

Ensuring sampled data reflects true trends

User Trust

DBMS Limitations

Benchmarking Rigor

02

QUERY COMPATIBILITY

Supporting complex SQL queries (e.g., joins, nested subqueries) without breaking approximation logic

03

PERFORMANCE TUNING

Balancing sampling rate and query latency to meet the 3× speedup goal

P.S.OVERVIEW

OUR KEY DELIVERABLES

01

WORKING PROTOTYPE

A query engine (standalone or DB-integrated) that supports approximate analytics.

02

DATASET TYPES

Assess both online and offline dataset to process out the pipeline in each case with remarkable improvements

03

SQL QUERY SUPPORT

Must handle basic analytical queries like COUNT, SUM, AVG, and GROUP BY

04

DOCUMENTATION

sampling, sketches, error bounds, and query rewriting.

How Our Solution Works?

Parse Query

- Detect APPROX, ERROR, PROBABILITY hints.
- Decide: Exact vs Approximate mode.

Plan Selection

- If Approximate: generate candidate sampling plans.
- Use DB's cost model (EXPLAIN) to choose efficient plan.

Pilot Sampling (Online)

- Run a small pilot query to estimate mean & variance.
- Check if error/confidence constraints can be met.

Main Sampling Execution

- Execute rewritten query on chosen sample.
- Aggregate results with block sampling adjustments.

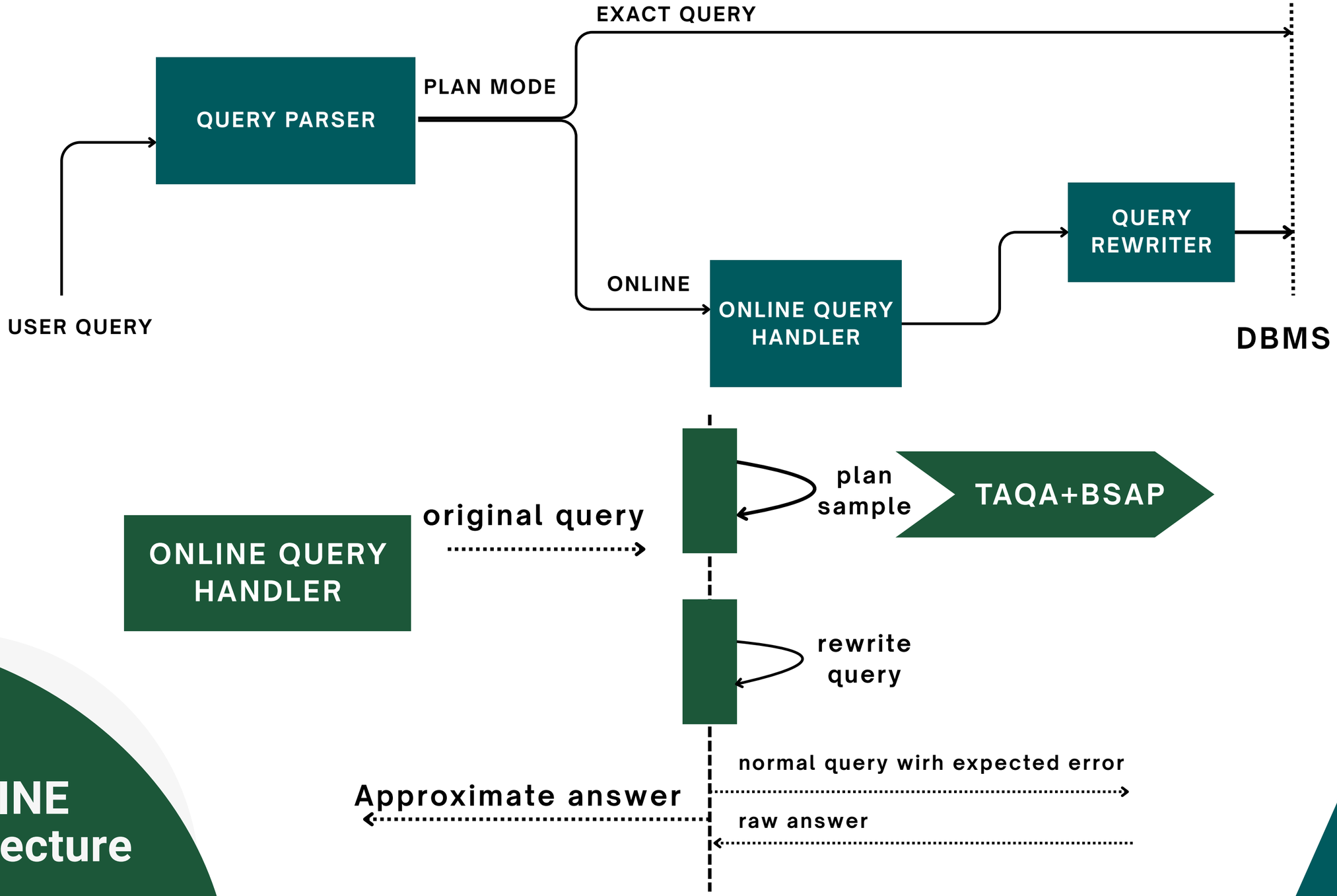
Post-Processing

- Compute unbiased estimate & confidence interval.
- Return results in same SQL format (transparent to user).



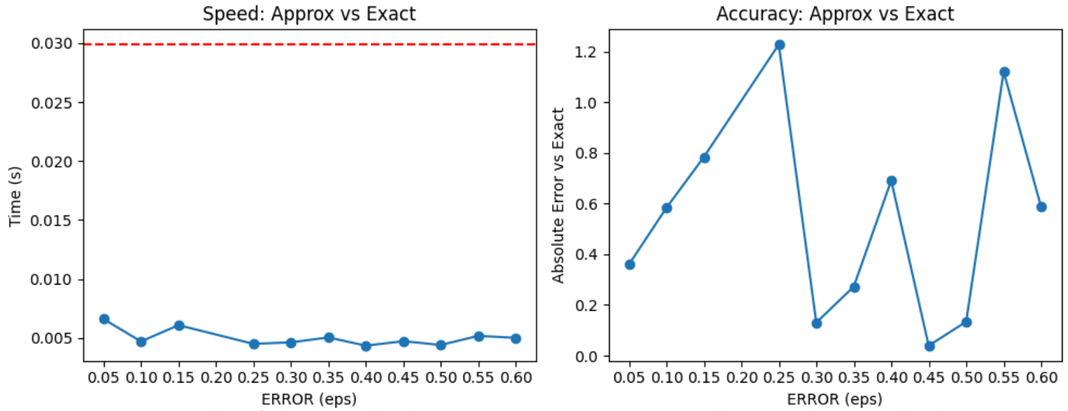
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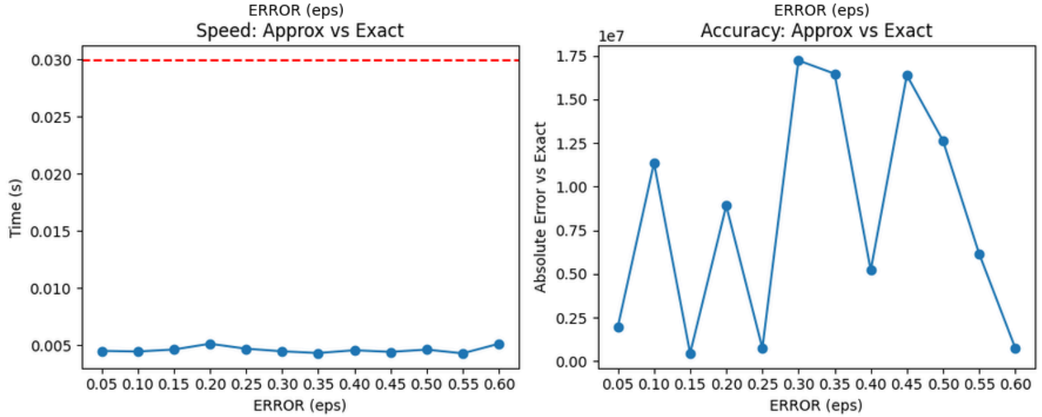
- https://arxiv.org/pdf/2503.21087v1
- https://arxiv.org/pdf/1804.00770
- https://arxiv.org/pdf/1203.5485

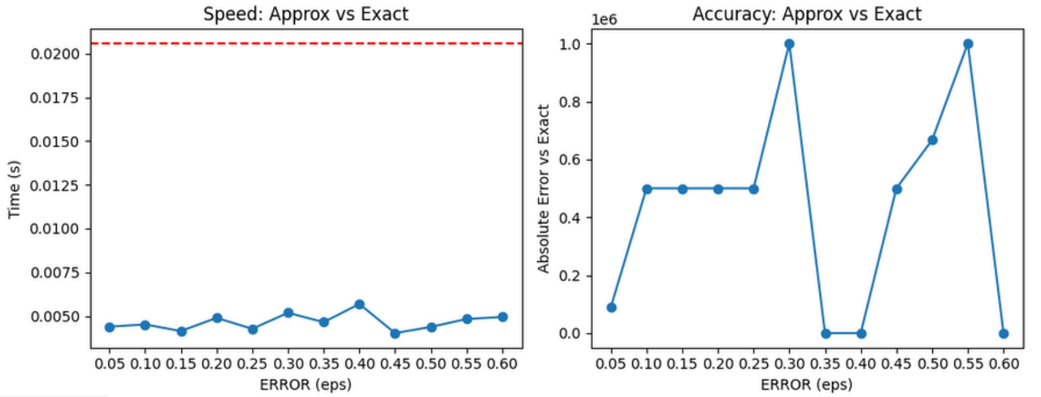


PIPELINE Architecture

BENCHMARK







ANALYSIS

1. AVG(Price) query

- Speed: Approx query runs much faster than exact (flat line ~0.005–0.01s) vs exact (red dashed, ~0.025–0.03s). The speedup is stable regardless of requested error tolerance.
- Accuracy: Absolute error fluctuates but stays relatively small (<1.2). The error doesn't strictly decrease with tighter ERROR specs — sometimes jumps due to sampling randomness.

2. SUM query

- Speed: Approx execution again ~0.005–0.01s vs exact ~0.025–0.03s. Stable speedup.
- Accuracy: Error values are much larger (scale ~10^7), and fluctuate heavily with ERROR spec. Some points align with the spec (error is small at tighter specs), but there's variability.

3. COUNT query

- Speed: Approx ~0.005–0.01s vs exact ~0.02s. Stable speedup.
- Accuracy: Errors are smaller (scale ~10^6 vs 10^7 for SUM), but also fluctuate strongly. At some error specs, the approximation is very close; at others, deviation spikes.

FEATURES

ASSUMPTIONS

- Queries are aggregation-heavy (e.g., SUM, AVG, COUNT)
- Data distribution is reasonably stable during sampling
- Users accept small approximation error with confidence bounds
- Database provides readable metadata (cost model, stats) for planning

LIMITATIONS

- Limited support for DISTINCT, LIMIT, and OFFSET in approximate mode
- Focused on single-table queries (joins not yet supported)
- Overhead makes it less efficient for small or very fast queries
- Requires live online sampling (offline precomputation removed)

ANALYSIS

FUTURE SCOPE

- Multi-Language Support → Parse queries from many languages into Postgres.
- Broader DB Support → Extend pipeline to offline DBs & other backends.
- Offline Mode → Allow local execution, less dependency on online DBs.
- Performance Optimization → Reduce lag from network calls vs. fast local queries.

FEATURES

- Dual Access: Integrated UI + CLI for flexibility
- Adaptive sampling with pilot + main phase execution
- Real-time error estimation with confidence intervals
- Lightweight wrapper API for easy catalog/database integration

APPLICATIONS

Speed at Scale

- Works even on billion-row or petabyte datasets.
- Delivers 2-3x faster insights for dashboards, monitoring, and analytics.

Adaptive Accuracy

- Users can tune error & confidence levels (e.g., ERROR 3%, PROB 95%).
- Ensures trustworthy approximations tailored to business needs.

Versatility Across Workloads

- Supports batch queries, ad-hoc analysis, and real-time streaming data.
- Applicable across BI, IoT, finance, healthcare, cybersecurity, and cloud analytics.

Seamless Integration

- Built on top of existing SQL engines (Postgres).
- No need to replace infrastructure → easy adoption & cost savings.



Thank You!